

```
#include<stdio.h>
void main()
{
    float p,r,t;
    printf("give the value of principle amount:");
    scanf("%f",&p);
    printf("the value of rate is:");
    scanf("%f",&r);
    printf("the value of time period is:");
    scanf("%f",&t);
    float SI=p*r*t/100;
    printf("the simple interest is %f\n", SI);
}
```

```
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\1.SI\simpleintrest.exe"
give the value of principle amount:23000
the value of rate is:5
the value of time period is:7
the simple intrest is 8050.000000
```

```
#include<stdio.h>
void main()
{
    float a,b,c,d,e,f,g;

    printf("give the marks of 5 subjects separated by space: ");
    scanf("%f %f %f %f %f",&a,&b,&c,&d,&e);
    f=a+b+c+d+e;
    g=f/5;
    printf("the total is %f and percentage is %f",f,g);
}
```

```
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\2.Marks\total_and_percentage.exe"
give the marks of 5 subjects separated by space: 89 98 78 87 79
the total is 431.000000 and percentage is 86.199997
PS C:\MinGW\bin>
```

```
#include <stdio.h>

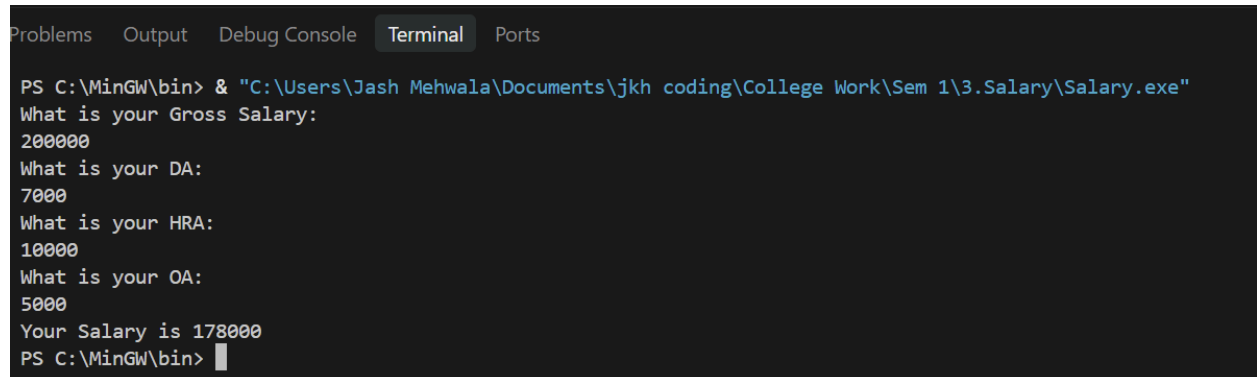
int main()
{
    int gs,da,hra,oa,s;
    printf("What is your Gross Salary:\n");
    scanf("%d",&gs);
    printf("What is your DA:\n");
    scanf("%d",&da);
```

```

printf("What is your HRA:\n");
scanf("%d",&hra);
printf("What is your OA:\n");
scanf("%d",&oa);
s=gs-da-hra-oa;
printf("Your Salary is %d",s);

}

```



```

Problems  Output  Debug Console  Terminal  Ports

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\3.Salary\Salary.exe"
What is your Gross Salary:
200000
What is your DA:
7000
What is your HRA:
10000
What is your OA:
5000
Your Salary is 178000
PS C:\MinGW\bin>

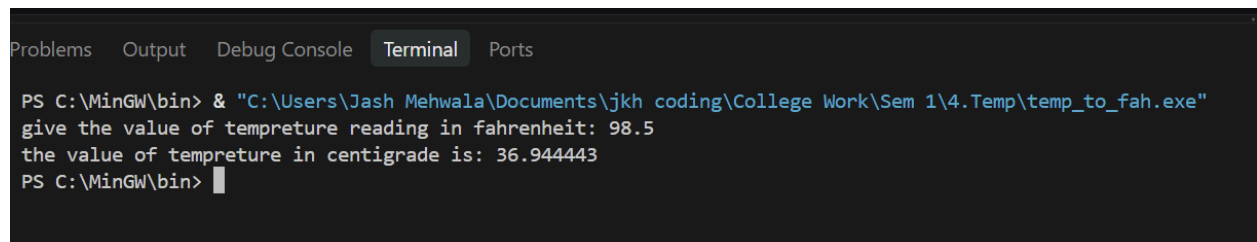
```

```

#include<stdio.h>
void main ()
{
    float t,f;
    printf("give the value of temperture reading in fahrenheit: ");
    scanf("%f",&f);
    t=((f-32)*100)/180;
    printf("the value of temperture in centigrade is: %f",t);

}

```



```

Problems  Output  Debug Console  Terminal  Ports

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\4.Temp\temp_to_fah.exe"
give the value of temperture reading in fahrenheit: 98.5
the value of temperture in centigrade is: 36.944443
PS C:\MinGW\bin>

```

```

#include<stdio.h>
void main()
{
    int a,b,c;
    printf("give the values of a and b respectively: ");
    scanf("%d%d",&a,&b);
}

```

```

printf("a is %d and b is %d before swap and \n",a,b);
c=a;
a=b;
b=c;
printf("a is %d and b is %d after swap",a,b);

}

```

The screenshot shows a Windows command prompt (MinGW) where a program named 'swap_with_3rd_var.exe' is executed. The user is prompted to enter values for 'a' and 'b'. The program outputs the values before and after the swap, confirming that the values have been correctly swapped using a third variable 'c'.

```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\5.Swap with\swap_with_3rd_var.exe"
give the values of a and b respectively: 23 32
a is 23 and b is 32 before swap and
a is 32 and b is 23 after swap
PS C:\MinGW\bin>

```

```

#include<stdio.h>
void main()
{
int a,b,c;
printf("give the values of variable a and b : ");
scanf("%d%d",&a,&b);
printf("the value of a and b before swap is %d and %d,\n",a,b);
a=a^b;
b=a^b;
a=a^b;
printf("the value of a and b after swap is %d and %d",a,b);

}

```

The screenshot shows a Windows command prompt (MinGW) where a program named 'swap_without_3rd_variable.exe' is executed. The user is prompted to enter values for 'a' and 'b'. The program outputs the values before and after the swap, confirming that the values have been correctly swapped using the XOR bitwise operation without a third variable.

```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\6 swap without\swap_without_3rd_variable.exe"
give the values of variable a and b : 12 23
the value of a and b before swap is 12 and 23,
the value of a and b after swap is 23 and 12
PS C:\MinGW\bin>

```

```

#include<stdio.h>
void main()
{
float b,h,a,i;
printf("give the value of base and height of triangle whose area we need to find\n");
printf("base: ");

```

```

scanf("%f",&b);
printf("height: ");
scanf("%f",&h);
a=0.5*h*b;
printf("The area of triangle is: %f",a);
/*
}
else
{
float A,B,C,s,area;
printf("give the values of sides A,B and C respectively:\n");
scanf("%f%f%f",&A,&B,&C);
s=(A+B+C)/2;
area=(s*(s-A)(s-B)(s-C));

}
*/
}

```

```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\7.Area of Triangle\area_of_triangle.exe"
give the value of base and height of triangle whose area we need to find
base: 23
height: 32
The area of triangle is: 368.000000
PS C:\MinGW\bin>

```

```

#include<stdio.h>
void main()
{
    char a;
    int i,d,h,m,s,i1=31558150,y=1,n=0;
    printf("Do you know the time in seconds in which the earth completes one revolution?(y/n)\n");
    scanf("%c",&a);
    /* if (a='y')
    {
        printf("Then enter the value");
        scanf("%d",&i);
        if(i==31558150)
        {
            printf("yes you are coreect! AND");
        }
    }
    */
}

```

```

    }
    else
    {
        printf("The time period of revolution of earth is 31558150 seconds AND ");
    }*/
    d=i1/(24*3600);
    h=(i1%(24*3600))/3600;
    m=(i1%3600)/60;
    s=i1%60;
    printf("the time of revolution of earth is %d days %d hours %d mins and %d seconds",d,h,m,s);
}

```

Problems Output Debug Console Terminal Ports

```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\8. Earth\Earth.exe"
Do you know the time in seconds in which the earth completes one revolution?(y/n)
n
the time of revolution of earth is 365 days 6 hours 9 mins and 10 seconds
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main()
{
    long int d,h,m,s,t;
    printf("Enter the time in Days,Hours,Minutes and Seconds: ");
    scanf("%ld%ld%ld%ld",&d,&h,&m,&s);
    t=(d*24*3600)+(h*3600)+(m*60)+s;
    printf("the total time in seconds is: %ld",t);

    return 0;

}

```

Problems Output Debug Console Terminal Ports

```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\9. Time_to_sec\timetosec.exe"
Enter the time in Days,Hours,Minutes and Seconds: 2 23 43 45
the total tim in seconds is: 258225
PS C:\MinGW\bin>

```

```

#include<stdio.h>
void main()
{

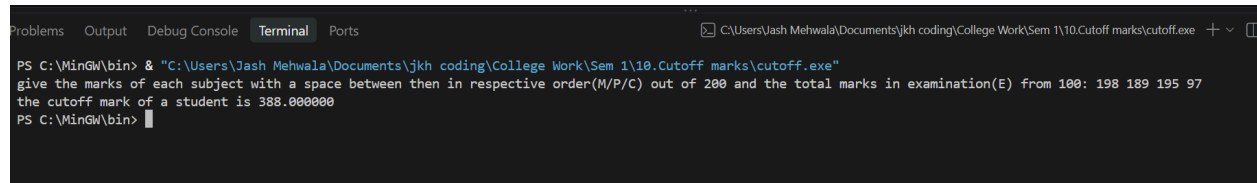
```

```

float CM,M,P,C,E;
printf("give the marks of each subject with a space between then in respective order(M/P/C) out
of 200 and the total marks in examination(E) from 100: ");
scanf("%f%f%f%f",&M,&P,&C,&E);
CM = M/2+P/2+C/2+E;
printf("the cutoff mark of a student is %f \n",CM);

}

```



```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\10.Cutoff marks\cutoff.exe"
give the marks of each subject with a space between then in respective order(M/P/C) out of 200 and the total marks in examination(E) from 100: 198 189 195 97
the cutoff mark of a student is 388.000000
PS C:\MinGW\bin>

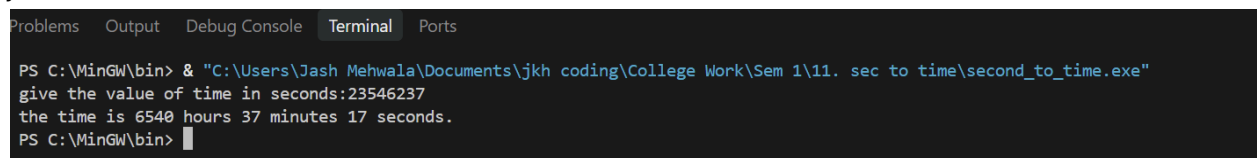
```

```

#include<stdio.h>
int main()
{
long int t,h;
int m,s;
printf("give the value of time in seconds:");
scanf("%ld",&t);
h=t/3600;
m=(t%3600)/60;
s=t%60;
printf("the time is %ld hours %ld minutes %d seconds.",h,m,s );

}

```



```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\11. sec to time\second_to_time.exe"
give the value of time in seconds:23546237
the time is 6540 hours 37 minutes 17 seconds.
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main ()
{
char c;
printf("Type the character whose ASCII code you want: ");
scanf("%c",&c);
printf("The char %c has ASCII code %d",c,c);

}

```

```
Problems Output Debug Console Terminal Ports
C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\12.Ascii\ASCII_Code_of_character.exe

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\12.Ascii\ASCII_Code_of_character.exe"
Type the character whose ASCII code you want: A
The char A has ASCII code 65
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
void main ()
{
int n, m;
printf("enter the number here: ");
scanf("%d",&n);
m= n%2;
if (m==0) {
    printf("your number is even");
}
else {
    printf("your number is odd");
}
}
```

```
Problems Output Debug Console Terminal Ports
C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\13.Even odd\even_or_odd.exe

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\13.Even odd\even_or_odd.exe"
enter the number here: 65
your number is odd
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main()
{
    int c;
    char ch;
    printf("type the character whose type you need\n");
    scanf("%c",&ch);
    c=ch;
    printf("%d\n",c);
    if (c==32)
    {
        printf("The character is the SPACE character");
    }
    else if (c>=48 && c<=57)
    {
        printf("The character is a NUMBER");
    }
}
```

```

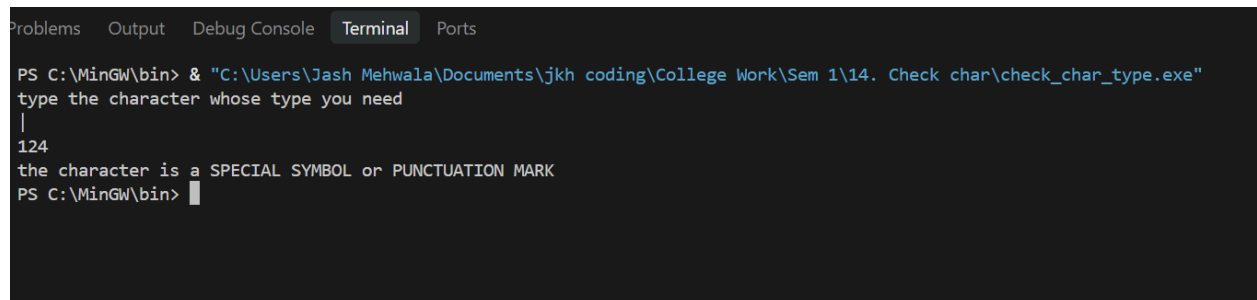
else if( c>=65 && c<=90)
{
    printf("the character is a CAPITAL LETTER");

}
else if( c>=97 && c<=122)
{

    printf("The character is a SMALL LETTER");
}

else{
    printf("the character is a SPECIAL SYMBOL or PUNCTUATION MARK");
}

```



```

Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\14. Check char\check_char_type.exe"
type the character whose type you need
|
124
the character is a SPECIAL SYMBOL or PUNCTUATION MARK
PS C:\MinGW\bin>

```

```

}
#include<stdio.h>
int main()
{
    int m;
    printf("Enter your marks\n");
    scanf("%d",&m);

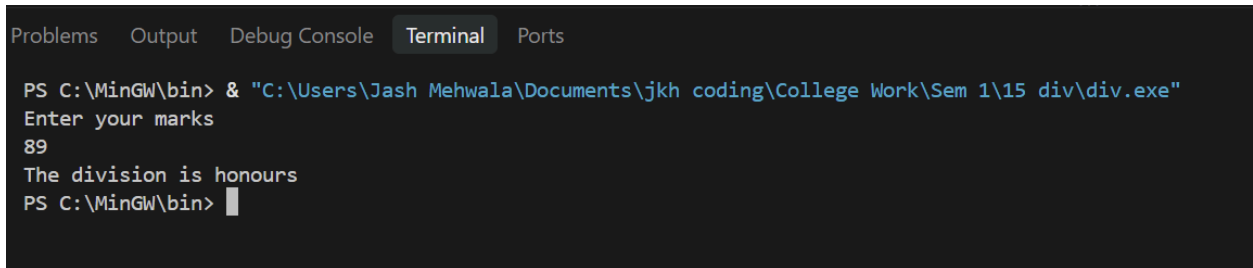
    if(m>=70 && m<=100)
    {
        printf("The division is honours");
    }
    if(m>=60 && m<70)
    {
        printf("The division is 1st Division");
    }
    if(m>=40 && m<60)
    {
        printf("The division is 2nd");
    }
    if(m>=30 && m<40)
    {

```

```

        printf("The division is 3rd");
    }
    if(m>=0 && m<30)
    {
        printf("Fail ");
    }
}

```



```

Problems  Output  Debug Console  Terminal  Ports

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\15 div\div.exe"
Enter your marks
89
The division is honours
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main()
{
    int a,b,c;
    printf("Give the value of three numbersw you want to compare with space between them\n");
    scanf("%d%d%d",&a,&b,&c);
    if(a>b)
    {
        if(a>c){
            printf("%d is the greatest number out of them",a);
        }
        else{
            printf("%d is the greatest of them",c);
        }
    }
    else{
        { if(b>c)
        {
            printf("%d is greatest of them",b);
        }
        else
        {
            printf("%d is the greatest number of the three\n",c);
        }
        }
    }
}

```

Problems Output Debug Console **Terminal** Ports

```
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\16.Max\max_of_3_numbers.exe"
Give the value of three numbersw you want to compare with space between them
45 54 34
54 is greatest of them
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main()
{
    int a,b,c;
    printf("Give the value of three numbersw you want to compare with space between them\n");
    scanf("%d%d%d",&a,&b,&c);
    if(a>b && a>c)
    {
        printf("%d is the greatest number out of them",a);
    }
    else if(b>c)
    {
        printf("%d is the greatest out of three", b);
    }
    else{
    {
        printf("%d is the greatest number of the three",c);
    }
    }
}
```

Problems Output Debug Console **Terminal** Ports

```
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\17.Max log\max_logic.exe"
Give the value of three numbersw you want to compare with space between them
23 45 78
78 is the greatest number of the three
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main()
{
    int a,b,c,m1;
    printf("Give the values of a,b and c respectively: ");
    scanf("%d%d%d",&a,&b,&c);
    m1=a>b?(a>c?a:c):(b>c?b:c);
    printf("The greatest number of them is %d\n",m1);
}
```

```
}
```

```
Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\18.Max Cond\Max_Cond.exe"
Give the values of a,b and c respectively: 65 74 53
The greatest number of them is 74
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main()
{
    char c,b;
    printf("give the character whose type you need to know\n");
    scanf("%c",&c);
    (c>='A' && c<='Z')?printf("%c character is a Capital letter\n",c):printf("%c is a Small
character\n",c);

}
```

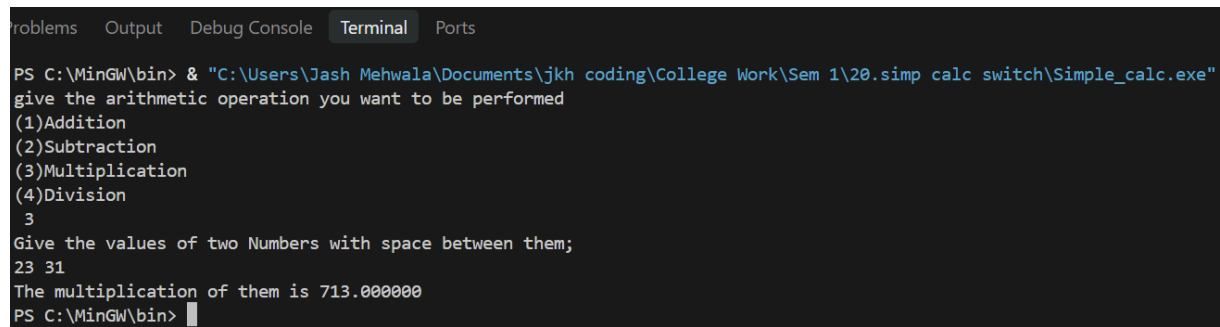
```
Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\19.Check char small usin cond\Char_usin_cond.exe"
give the character whose type you need to know
A
A character is a Capital letter
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main()
{
    int a;
    float b,c,d;
    printf("give the arithmetic operation you want to be
performed\n(1)Addition\n(2)Subtraction\n(3)Multiplication\n(4)Division\n ");
    scanf("%d",&a);
    printf("Give the values of two Numbers with space between them;\n");
    scanf("%f%f",&b,&c);
    switch(a){
        case(1):
            d=b+c;
            printf("The addition of them is %f",d);
            break;
        case(2):
            d=b-c;
            printf("The subtraction of them is %f",d);
            break;
        case(3):
            d=b*c;
            printf("The multiplication of them is %f",d);
```

```

        break;
    case(4):
        d=b/c;
        printf("The division of them is %f",d);
        break;
}
}

```



```

Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\20.simp calc switch\Simple_calc.exe"
give the arithmetic operation you want to be performed
(1)Addition
(2)Subtraction
(3)Multiplication
(4)Division
3
Give the values of two Numbers with space between them;
23 31
The multiplication of them is 713.000000
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main()
{
    char a;
    float b,c,d;
    printf("give the arithmetic operation you want to be
    performed\n(+)Addition\n(-)Subtraction\n(X)Multiplication\n(/)Division\n ");
    scanf("%c",&a);
    printf("Give the values of two Numbers with space between them;\n");
    scanf("%f%f",&b,&c);
    switch(a){
        case('+'):
            d=b+c;
            printf("The addition of them is %f",d);
            break;
        case('-'):
            d=b-c;
            printf("The subtraction of them is %f",d);
            break;
        case('X'):
            d=b*c;
            printf("The multiplication of them is %f",d);
            break;
    }
}

```

```

    case('/'):
    d=b/c;
    printf("The division of them is %f",d);
    break;
}
}

```

```

PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\21.20 but diff\diff.exe"
give the arithmetic operation you want to be performed
(+)Addition
(-)Subtraction
(X)Multiplication
(/)Division
X
Give the values of two Numbers with space between them;
12 21
The multiplication of them is 252.000000
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main()
{
    long int i,f=1,i1,k;
    printf("Give the number whoes factorial you want\n");
    scanf("%ld",&i);
    i1=i;
    for(k=i;k>0;k--)
    {
        f=f*k;
    }
    printf("The value of %ld! is %ld",i1,f);

}

```

```

Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\22. Factorial\Factorial.exe"
Give the number whoes factorial you want
4
The value of 4! is 24
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main()
{
    int i,f=1,i1,k;
    printf("Give the number whose power you want\n");
    scanf("%ld",&i);
    printf("The power it has to be raised to\n");
    scanf("%ld",&i1);
    for(k=i1;k>0;k--)

```

```

{
f=f*i;
}
printf("The answer is %d",f);

}

```

```

Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\23.Power\powe.exe"
Give the number whose power you want
45
The power it has to be raised to
4
The answer is 4100625
PS C:\MinGW\bin>

```

```

#include<stdio.h>
int main()
{
    int x,y,n;
    printf("Give the value of x and n\n");
    scanf("%d%d",&x,&n);
    if(n==1)
    {
        y=1+x;
    }
    else if (n==2)
    {
        y=1+(x/n);
    }
    else if (n==3)
    {
        int i,X=1;
        for(i=n;i>0;i--)
        {
            X=x*X;
        }
        y=1+X;
    }
    else
    {
        y=1+n*x;
    }
    printf("The value of y is %d",y);
}

```

```
}
```

```
Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\24. y\y.exe"
Give the value of x and n
3 4
The value of y is 13
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main()
{
int n,i,y;
printf("Enter the number whose value\n");
scanf("%d",&n);
for(i=1;i<11;i++)
{
y=n*i;
printf("%d X %d = %d\n",n,i,y);
}
}
```

```
Problems Output Debug Console Terminal Ports
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\25.Table\table.exe"
Enter the number whose value
23
23 X 1 = 23
23 X 2 = 46
23 X 3 = 69
23 X 4 = 92
23 X 5 = 115
23 X 6 = 138
23 X 7 = 161
23 X 8 = 184
23 X 9 = 207
23 X 10 = 230
PS C:\MinGW\bin> █
```

```
#include<stdio.h>
int main ()
{
```

```

int n,o=0,e=0,i,j;
printf("Enter the value of n\n");
scanf("%d",&n);
    for(i=0;i<=n;i=i+2)
    {
        e=e+i;
    }
    for(j=1;j<=n;j=j+2)
    {
        o=o+j;
    }
printf("The sum of odd and even numbers btw 0 to %d is %d and %d",n,o,e);

}

```

```

The sum of odd and even numbers btw 0 to N is 9 and 6
PS C:\MinGW\bin> & "C:\Users\Jash Mehwal\Documents\jkh coding\College Work\Sem 1\26. Sum odd even\sum.exe"
Enter the value of n
89
The sum of odd and even numbers btw 0 to 89 is 2025 and 1980

```